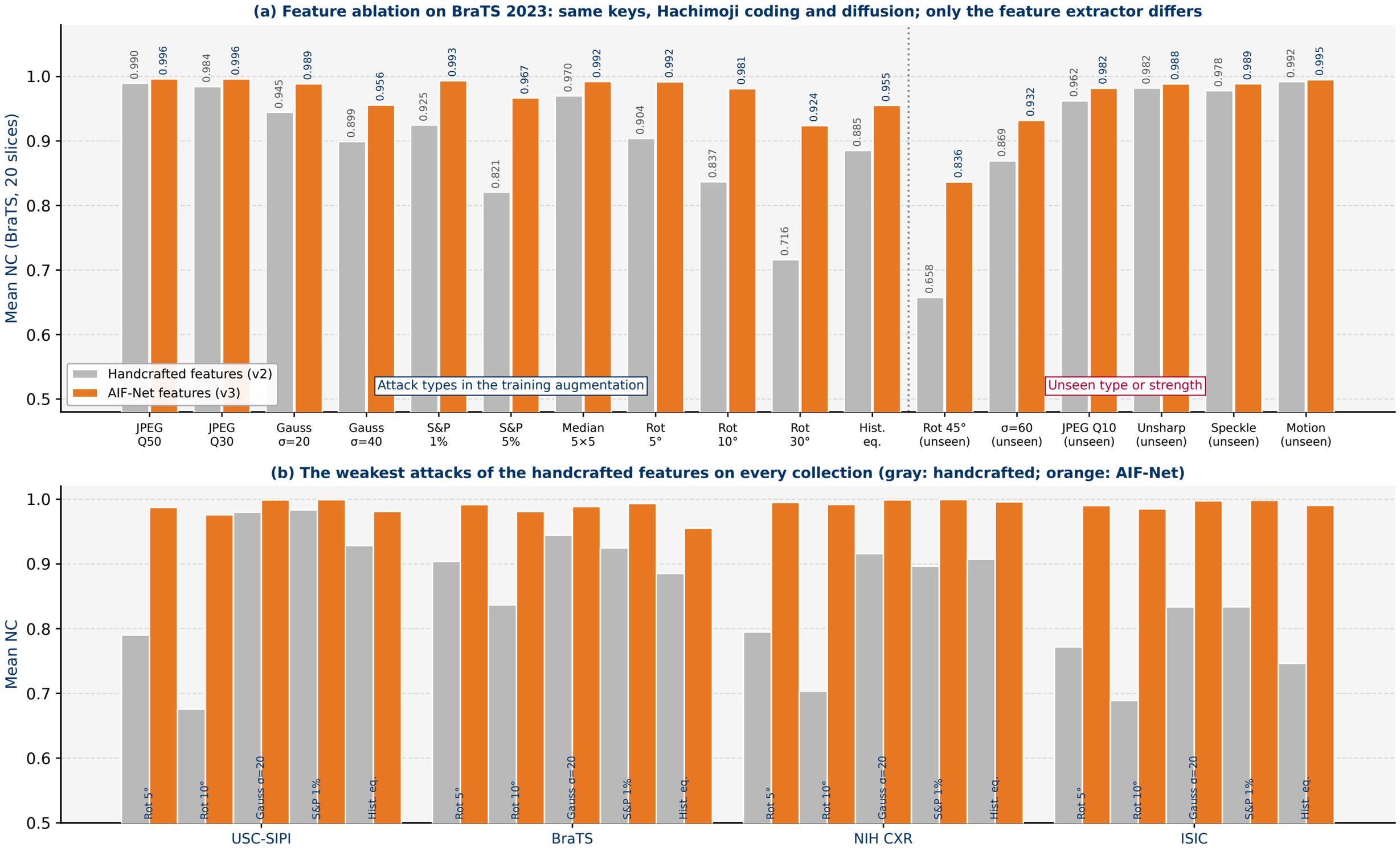
(a) Feature ablation on BraTS 2023: same keys, Hachimoji coding and diffusion; only the feature extractor differs
Mean NC (BraTS, 20 slices)
1.0
0.9
0.8
0.7
0.6
0.5
Handcrafted features (v2)
AIF-Net features (v3)
Attack types in the training augmentation
Unseen type or strength
0.990
0.996
0.984
0.996
0.945
0.989
0.899
0.956
0.925
0.993
0.821
0.967
0.970
0.992
0.904
0.992
0.837
0.981
0.716
0.924
0.885
0.955
0.658
0.836
0.869
0.932
0.962
0.982
0.982
0.988
0.978
0.989
0.992
0.995
JPEG Q50
JPEG Q30
Gauss σ=20
Gauss σ=40
S&P 1%
S&P 5%
Median 5×5
Rot 5°
Rot 10°
Rot 30°
Hist. eq.
Rot 45° (unseen)
σ=60 (unseen)
JPEG Q10 (unseen)
Unsharp (unseen)
Speckle (unseen)
Motion (unseen)
(b) The weakest attacks of the handcrafted features on every collection (gray: handcrafted; orange: AIF-Net)
Mean NC
1.0
0.9
0.8
0.7
0.6
0.5
Rot 5°
Rot 10°
Gauss σ=20
S&P 1%
Hist. eq.
USC-SIPI
BraTS
NIH CXR
ISIC